This document is intended to provide a basis for Midterm Senior Project Evaluations and is intended for use by the Faculty Coach, the Project Team, and any stakeholders, including the Project Sponsor. The evaluation criteria below are divided into two main areas: Process/Project, and Product. Early in the project, the primary focus was placed on the Process/Project areas; in this evaluation we will focus much more heavily on product. Because each area is expansive and often includes entire classes covering them, a sample rubric, some questions, and general evaluation categories have been provided to increase understanding of what is expected throughout the project.

To use this document, go through each area and facilitate discussion. It is not necessary to complete each rubric or to fill in each question, but students should be encouraged to individually do so, meet as a group for comparison, and report back to the faculty coach and sponsor any results or conclusions they reach.

Process and Project

The tendency to focus on the product often leaves the process and the project out in the cold, especially in projects with set or compressed schedules. A well developed Software Engineering team knows that the returns paid within the Process and Project domains far outweigh the time and expenses that occur. Although the list below is not exhaustive, it includes some of the primary areas of importance.

Meetings and Communications Management

Sample Rubric: Meetings and Communications Management				
Meetings were held only at regular intervals or as a result of a crisis and did not include agendas or pre-planning	Meetings included standing agendas which were occasionally followed	Meetings included predetermined agendas, but the agenda was not used within the meeting or evaluated afterward. Included some items which did not necessitate a meeting	Meetings included predetermined agendas and were somewhat organized, but were occasionally allowed to stray from format. Content necessitated a meeting	Meetings were pre-organized, included prepared agendas, followed expected format, and did not include content that did not necessitate a meeting
Communication with stakeholders was reactive or crisis-driven, did not occur when expected, and did not clearly communicate expectations	Communication was conducted out of necessity but was rarely proactive and did not account for stakeholder expectation management	Communication was scheduled and typically took place. Expectation management was rarely contemplated	Communications with stakeholders were regular and generally proactive. Expectation management was occasionally used.	Communications with stakeholders were regular, proactive, and purpose-driven. Language was clear and included expectation management
In terms of customer education, team had a negative attitude towards the customer	Team blindly accepted customer inputs, did not convey own knowledge adequately	Team missed some occasions when customers needed education or	Team provided technical inputs to customer when needed or asked for	Team made sure at every stage to educate customer and provide/seek clarifications

and did not bother to assist them		clarifications were needed		

Comments:
The team has been on track for sponsor meetings.

Thursday team meetings have been structured based on agendas and standups.

Decisions are jointly made between the team and sponsor

1. Is the team conducting effective meetings? What can be changed to make the team meetings more productive?
   .
   Sponsor meetings sometimes go off track. The team would like to make it clear that when sponsor meetings end early we plan on using the remaining time for work activities such as reviewing sprint progress, quick standups, communicating bugs/issues etc…

2. How well has the team been communicating project progress to the sponsor? What regular communication does the team have with the sponsor? Has the team been maintaining this communication to the satisfaction of the sponsor? Were any adjustments needed in the communication over time? Were these changes initiated by the team or the sponsor?

   We've made active improvements to keep the sponsor as up-to-date as possible with the project progress. Other than our weekly meetings, the whole team communicates with him through Discord for any needs. This communication has been consistent since adding him to the channel. The aforementioned change has been a huge boon for us, and was initiated by the team.

Sponsor/Stakeholder Communications

Comments:
Meetings are typically efficient and cover necessary content, they are focused and purposeful

Discord is also used to discuss urgent topics with the sponsor

Time and Effort

Sample Rubric: Time and Effort				
Student effort was minimal, not meeting minimum	Students effort only met minimum standards, and	Student effort was adequate to meet project requirements,	Student effort was consistently better than average, even	Students consistently put in full effort, even when internal or

requirements and putting the success of the project in crisis	excuses or blame was often used to explain the effect of internal or external hampering factors	but was frequently hampered by both internal and external factors	with negative internal factors. External factors occasionally affected effort levels	external factors have hampered progress
Comments: We are definitely over the line.				

1. Has the level of effort of individual team members been reflected in the output of the team as a whole?

   Despite detrimental external factors, the team has ramped up progress since the beginning of the semester. The output of the team has been mostly similar between team members, with certain weeks having more output for certain members than others.

2. To what extent has thrashing (unproductive work) become a factor?

   There has been some thrashing due to problems deploying our app to the Shopify App Store. Transferring ownership of the app from our organization to the Sponsor's was not a smooth process both in part due to poor documentation for our exact issue as well as our own inexperience. Ultimately, the team needed to manually change the config of the code under a new app in the sponsor's organization, a solution that is not necessarily endorsed by Shopify.

Conformance to Schedule

Sample Rubric: Conformance to Schedule				
Did not demonstrate any schedule management and did not meet schedule requirements	Met minimal schedule requirements but showed no schedule management	Met most schedule requirements with little or no active schedule management	Met most or all schedule requirements with some active schedule management	Actively managed and met all schedule requirements
Comments: The team has missed some important commitments but remains dedicated to meeting the specified requirements.				

1. Has the team met all project milestones to date? Which milestones, if any, were missed or were met ahead of schedule? What contributed to this schedule changes? What will the team do differently to ensure that future milestones are met?

   Deployment is still slightly delayed due to processes that could only be completed after the app has changed ownership. For example, we need to meet metrics calculated by Shopify in order to distribute the app; however, these calculations are not initiated until the app listing is ready. Going forward the team will ensure that beta testing feedback is captured properly and acted upon in a timely manner to maximize the effect it can have on the product.

Change and Scope Management

Sample Rubric: Change and Scope Management				
Did not demonstrate awareness of issues as change or scope issues but rather as individual problems	Occasionally identified change or scope issues as such but often addressed them reactively as individual problems	Addressed change and scope when issues were presented with little or no root cause analysis	Addressed change and scope regularly but not always actively addressing change and scope issues before they occurred	Actively managed change and scope, seeking out and addressing issues before they affected the project. Used root cause analysis
Comments: The sponsor believes the current scope of the project is clear.				

1. Has the scope of the project been considered at each milestone?

   At the beginning of this semester the team spent much consideration on the direction and goals of the project for this iteration. We have decided to place deployment and beta feedback at the highest level of priority with maintenance tasks such as refactoring and testing coming next. Adding additional features is a possibility however, will be time permitting only. These measures will ensure that the resultant product handed off to our sponsor is of the highest quality and meets the minimum set of requirements outlined during the kick-off.

Risk Management

Comments:

Deliverables

Sample Rubric: Deliverables				
Deliverable was not present and not substituted with other efforts or deliverables	Deliverable was delivered with minimal effort and did not contribute significantly to the success of the project. Some obvious problems or inconsistencies were present, demonstrating little QA occurred	Deliverable met standards and was useful at introduction, but was not updated or useful further along in the project. Few obvious problems or inconsistencies were present	Deliverable was better than average and was used or enhanced irregularly, performing its desired function within the project. No obvious problems or inconsistencies were present	Deliverable performed as expected, was reviewed and enhanced at regular intervals, and would have consistently allowed someone not familiar with the project to become informed
Comments: Add documents to git and keep them updated				

1. How often do these artifacts get updated on the department project website?
   All artifacts that should be displayed on the portal are there, but design documents have not been updated as frequently as they could be. This is due to our focus on deployment.

2. Has the team had any issues with configuration management? How were these problems solved? What percentage of project artifacts is under configuration control?
   Yes, mainly due to wanting to get the beta deployment out, but we've implemented a more thorough process in place to ensure that new documents are created properly when needed, and those that need updates are noted down and have a task assigned to tackle them rather than just trying to remember that we have to do it.

Project Website

Comments:

Risk Management Table

Comments:

Quality

Sample Rubric: Quality				
Little attention was paid to initial requirements and the final product related only loosely to them	Requirements were created and used as a guideline for initial efforts, but were rarely addressed later in the project	Requirements management and quality control were performed irregularly, but there was an understanding of the link between quality and conformance to requirements	Conformance to functional requirements was a priority, but non-functional requirements often took a back seat, relating to quality control	Conformance to functional and non-functional requirements was clearly a priority and actively prevented scope creep and similar issues
A customer-view of quality was not considered throughout the project	Customer satisfaction was implied as a requirement but not effectively understood by the team. Treated as a requirement rather than a separate quality factor	The group understood that customer satisfaction was important, and occasionally addressed it, but focused primarily on requirements	Customer satisfaction was a priority, and some attention was placed to areas such as maintainability, but some imbalance occurred	Customer satisfaction and a long-term scope as a priority was demonstrated in each project step and properly balanced with conformance to requirements and scope
Comments: The coach and sponsor see this area as positive.				

1. Has the work accomplished to date satisfied the project sponsor? Team's opinion? Sponsor's opinion? Were there any weak spots in this regard?

   The sponsor has given us positive feedback on the work we have accomplished so far. The team wishes we would have gotten farther by now in terms of features completed, but we are not disappointed in our current progress. There is a slight delay in beta deployment that changed our timeline, but we are working hard to deal with this.

Process Creation and Improvement

Sample Rubric: Process Creation and Improvement				
Process is only used when clear best-practice standards exist or an external process must be followed	Processes have been created and are usually followed, but are rarely improved.	Processes have been created and are consistently followed, but are rarely improved. Crisis situations occasionally trigger process improvement	Processes are always followed and occasionally improved. Root cause analysis may be a byproduct but is not actively pursued	Active process improvement is addressed regularly and processes are evaluated/root cause analysis is used regularly

Metrics were not planned or used throughout the project	Metrics were used only as required by faculty coach or project guidelines, but were rarely used and only tracked by necessity	Metrics were regularly collected, but inconsistently used, especially relating to process	Metrics were regularly collected and evaluated/used as intended in process and product domains.	Metrics were properly planned, implemented, proved useful to stakeholders, and were used for process improvement
Comments: The coach has not seen much in the area of metrics.				

1. What is the set of metrics that the team is tracking? Has the team gathered these metrics on a consistent basis? What has the team learned from the review of these metrics?

   The team is tracking the following metrics currently:

   a. **Effort By Type**

      i. **We have diagramed effort by type charts for our sprints and it helps us visualize our activity distribution in different categories (development, design, etc.). Analyzing these charts has helped us during our sprint planning allowing us to evenly distribute our work across different categories.**

   b. **Sprint Velocity**

      i. **As we monitored sprint velocity last semester we noticed that the amount of points completed each sprint steadily increased. This is due in part to the fact that many tickets were 8 or more points but also because the team began working harder as deadlines approached. We are addressing this situation by writing smaller tickets - an action that is only made possible because we have more maintenance to be done then the creation of features.**

   c. **This semester the team will calculate SUMI scores based on feedback from beta testers of the app**

      i. **SUMI scores will be calculated from our SUMI form that asks a set of questions based on software usability goals. From these scores the team can better understand the usability of the app and make adjustments as necessary.**

Team Effectiveness

Sample Rubric: Team Effectiveness				
Team fractured, unable to work	Team had some significant frictions	Team had some significant disconnects	Team worked reasonably well	Team has achieved synergy, allocated work well, and

together constructively.	and lack of mutual cooperation.	and frictions, but worked through them.	together, no problems visible.	worked together as a cohesive whole.
Comments: The coach thinks we seem to be an effective team.				

1. Is this an effective team? What has been contributing to and detracting from the team's effectiveness? What are the team's weak points? What are the team's strong points? What changes can the team make for next term that will make it more effective?

   Yes, this is an effective team. We have been able to synergize our efforts in creating our product. For example, many features intertwine with each other and as such team members have effectively communicated across product boundaries in order to successfully produce working code. We do well to ensure that all members are on the same page about the current state of the product and where we are heading next. One area the team could improve on is to respond to questions more frequently, even if someone has already answered. This applies to pull requests.

Product

Although Process and Project are important, the product is ultimately why projects are initiated and followed through to completion. An effective Software Engineer maintains a steady focus on the product, and understands the scope of what it entails. Although the list below is not exhaustive, it includes some of the primary areas of importance, especially for the second part of the project.

Design

Sample Rubric: Requirements				
Hardly any design done or documented.	Design significantly inappropriate and missed several key concerns.	Design somewhat inadequate – some concerns not addressed or documented adequately	Design adequate and appropriate and documented. Major design concerns addressed.	Several alternatives explored, eventual design well-suited to application needs.
Comments: We need to make sure we have the non-functional requirements for designs and documentation.				

1. Products need to be designed within guidelines and constraints appropriate for each project. It is also important to consider the impacts of the products that are designed. In the following categories discuss the constraints and impacts that have a bearing on your project. Note that there may be one or two categories that have no bearing on your project but your project is probably affected by almost all of these.
    a. Economic issues

    <u>Constraint</u>: In times of economic hardship our product has a high chance of being used far less as people would be looking to save money, and subscriptions are likely some of the first things that people will stop buying.
    In addition, the application will only be deployed to the shopify app store. This means that only merchants who use the Shopify platform can be considered potential customers.

    <u>Impact</u>: Improper implementation can lead to wasting company time and money. At the very least our project has to provide value to companies with the subscription management and analytics, if not then that serves as an issue.

    b. Environmental issues

    <u>Constraint</u>: If there are environmental issues such as natural disasters, it can bring servers down leading to businesses losing money.

<u>Impact</u>: Utilizing AI and servers to host and run business raises concerns with resources consumptions. Servers can consume large amounts of energy which can have negative impacts on the environment.

Social issues
<u>Impact</u>: Our project can be viewed as slightly anti-consumer since it adds another layer before they can fully cancel their subscription, but at the same time we offer and do provide discounts.

c. Political issues
<u>Constraint</u>: AI regulations may affect the ChurnGuard AI model. Legislation has previously been considered to enforce merchants to make it just as easy to sign up for a subscription plan as it is to cancel one.

<u>Impact:</u> At this point the team has not much considered political issues of the app, however, these will determine the future success. If merchants can no longer attempt to prevent cancellations then our app will fail. Regardless, our app aims to serve customers as well by offering them discounts. Even if disadvantageous legislation passes, our app may be viewed favorably for this reason.

d. Ethical issues
<u>Impact</u>: The scope of the project changed from emailing potential lost subscribers before they cancel to intervening when they attempt to cancel. Sending an email to an inactive customer may remind them of a subscription they forgot they had. Thus, the design is taking advantage of people forgetting they are subscribed. Additionally, getting in the way of customers canceling is not great from a consumer perspective.

<u>Constraint</u>: Since the main distinguishing feature of this service involves discounts, some customers may attempt taking advantage of the discount system. We need to prepare for this.

e. Health and safety
N/A

f. Manufacturability
<u>Constraint</u>: If a product isn't able to be produced quickly, or produced as quickly as the demand requires, then subscriptions for our product become hard to manage/impractical

<u>Impact</u>: Our project can help advertise certain products a merchant has, and if a certain one or many get very popular, the manufacturer may struggle keeping up with demand.

g. Sustainability

<u>Constraint</u>: Our project is only available if Shopify is available. Even though it is unlikely Shopify goes down, it's always a possibility. AWS constraint as well.

<u>Impact</u>: If the app is not user friendly then admins will have trouble creating and managing subscriptions.

Implementation

Sample Rubric: Requirements				
Implementation quality is completely unacceptable. Documentation poor	Implementation artifacts had significant problems or were difficult to understand; Implementation is not documented well.	Implementation is marginal or not well thought through. Implementation is sufficiently documented but is hard to understand.	Implementation artifacts are reasonably sound. Implementation is well documented and easy to understand.	Implementation artifacts of excellent quality Implementation is well documented and easy to understand
Comments: The team has rushed a bit on implementation for the purpose of getting features out.				

1. How has the development and implementation progressed? What percentage of the product do you estimate is complete at this point? Is the team providing the documentation within the implementation artifacts?

At this point almost all of the original requirements are fulfilled. The team is working to deploy an early beta version that will provide this functionality to merchants. Development was mainly focused on feature implementation last semester and has since shifted towards maintenance.

Testing

Sample Rubric: Requirements				
Hardly any testing done. An afterthought.	Testing was completely ad hoc and inadequate.	Test strategy inadequate or inappropriate, testing not thorough.	Testing was reasonably complete, strategy acceptable. Testing has been a major focus of the project.	Very sound strategy, thorough testing of all deliverables, including elements of test-driven development.
Comments: Same as before, the rushing has been at the expense of testing. The new recruit is focused on testing.				

1. What is the team's testing strategy? Has the team developed a test plan? Is the team performing unit testing? Is the team using any test frameworks, such as JUnit? What are the testing results to date? Were any major defects found during the system test?

The team has started implementing more automated tests and wants to begin checking core web vitals. The team has developed a test plan incorporating all the aforementioned methods as well as manual UI testing, and will soon start beta testing as well. Results are TBA, but from manual testing thus far we have found some defects, but those have all been mainly related to UI defects such as not displaying the right values, etc.

Quality: Verification and Validation

Sample Rubric: Requirements				
Key quality concerns not identified or addressed.	Patchy attention to quality concerns, some concerns missed.	Quality concerns addressed somewhat inadequately.	Reasonably good job of identifying and addressing quality concerns.	Quality concerns explicitly identified and considered fully at every stage.
Comments: The coach wouldn't give anyone a 5 on this, but agrees we have done well identifying issues.				

Quality Improvement Areas

Comments: Nothing new.

Future Considerations:

Final evaluation will weigh heavily on product quality, especially as it relates to customer satisfaction. What are the implications of these at the present point in the project?

At the current state of the project it seems by the end of the semester the team will have produced a fully working application with the proper documentation to be extended by future iterations. It is implied that the team will have completed all of the base functionality of the app with enough proof of concept for uses of AI that will allow the app to compete in its market space. That said, the app will likely not be fully "complete" or otherwise meet the entire original specifications that included extra features. The team has already managed this expectation to ensure that our final product is of the highest quality in its own right.